VICTORY 2027

The Nervous System

LECTURE 1 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Transmission of the Nerve Impulse
2. Types of Neurons

PWNSAT

WHO CAN APPEAR?
Classes 6th – 12th & Droppers

EXAM DATES

ONLINE EXAM
1st to 15th Oct (Daily)

OFFLINE EXAM
3rd, 4th, 10th & 11th Oct

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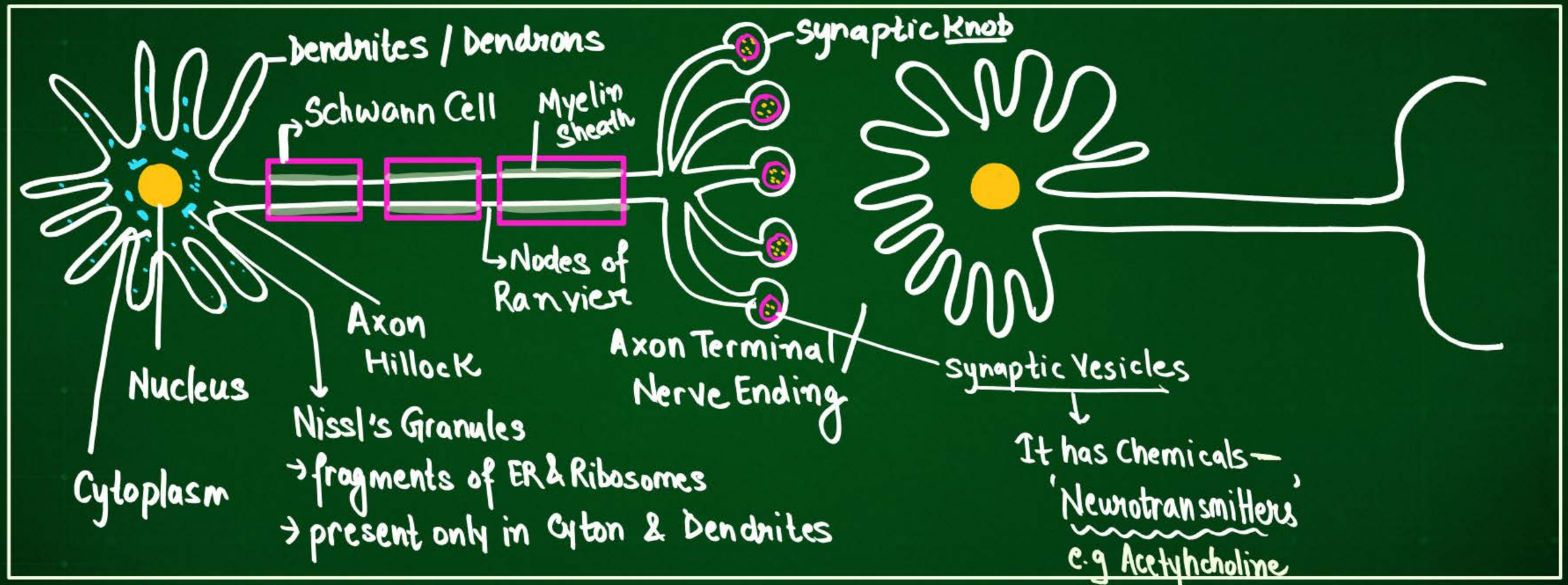
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'Neuron / Nerve Cell'

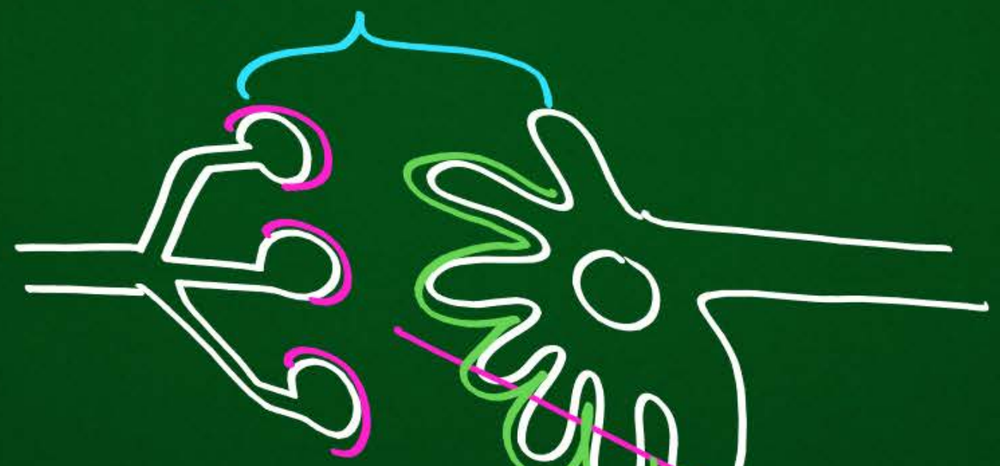
- No Centrosome/ Centriol
- longest
- Structural & functional unit of Nervous System

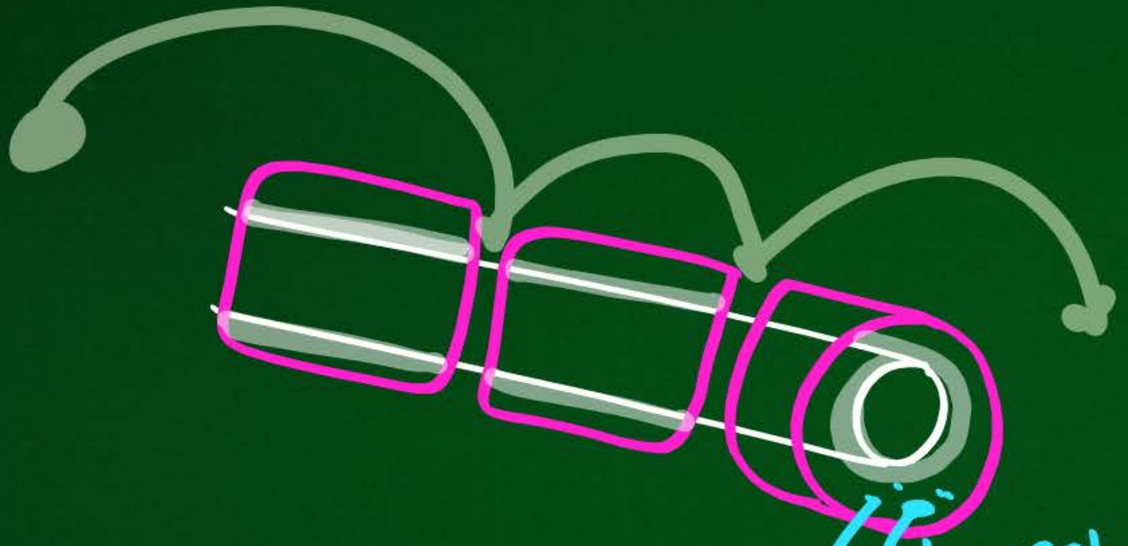


Cyton / Cell Body / Perikaryon → Cytoplasm + Nucleus

Synapse. → The microscopic gap between two adjacent neurons that is known as Synapse.

Synapse is divided into 3 parts -

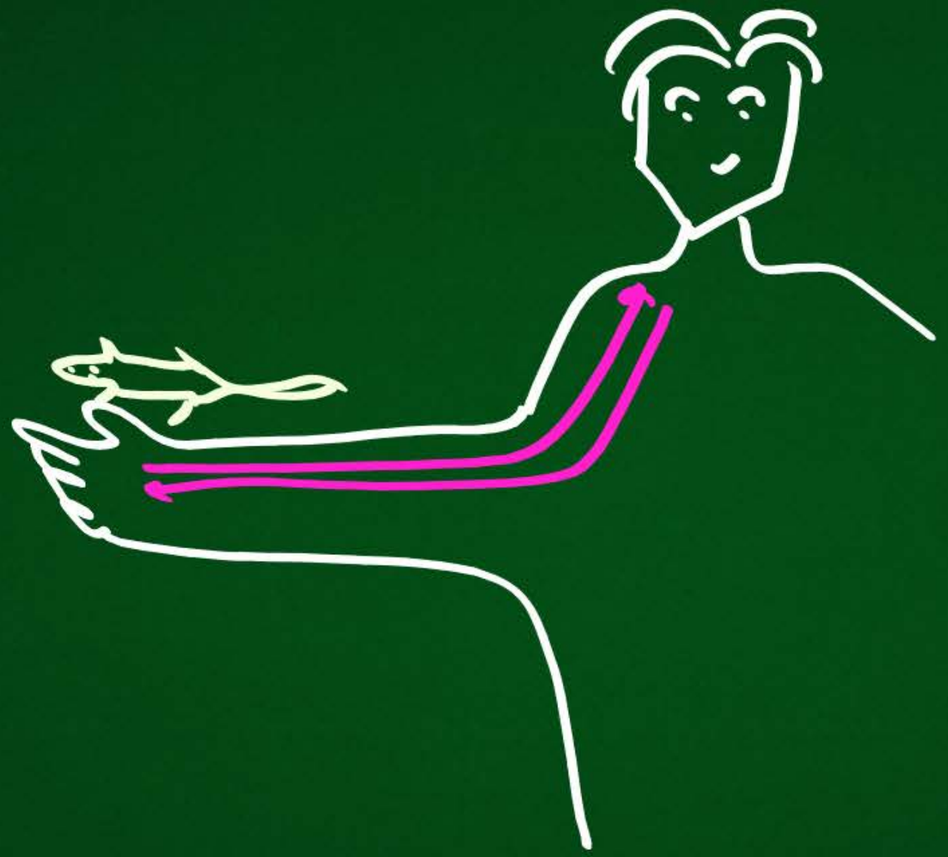




Neurilemma is the cell membrane, cytoplasm & Nucleus of Schwann Cell.

Schwann Cell
Cell Membrane
Nucleus
Cytoplasm

Neurilemma

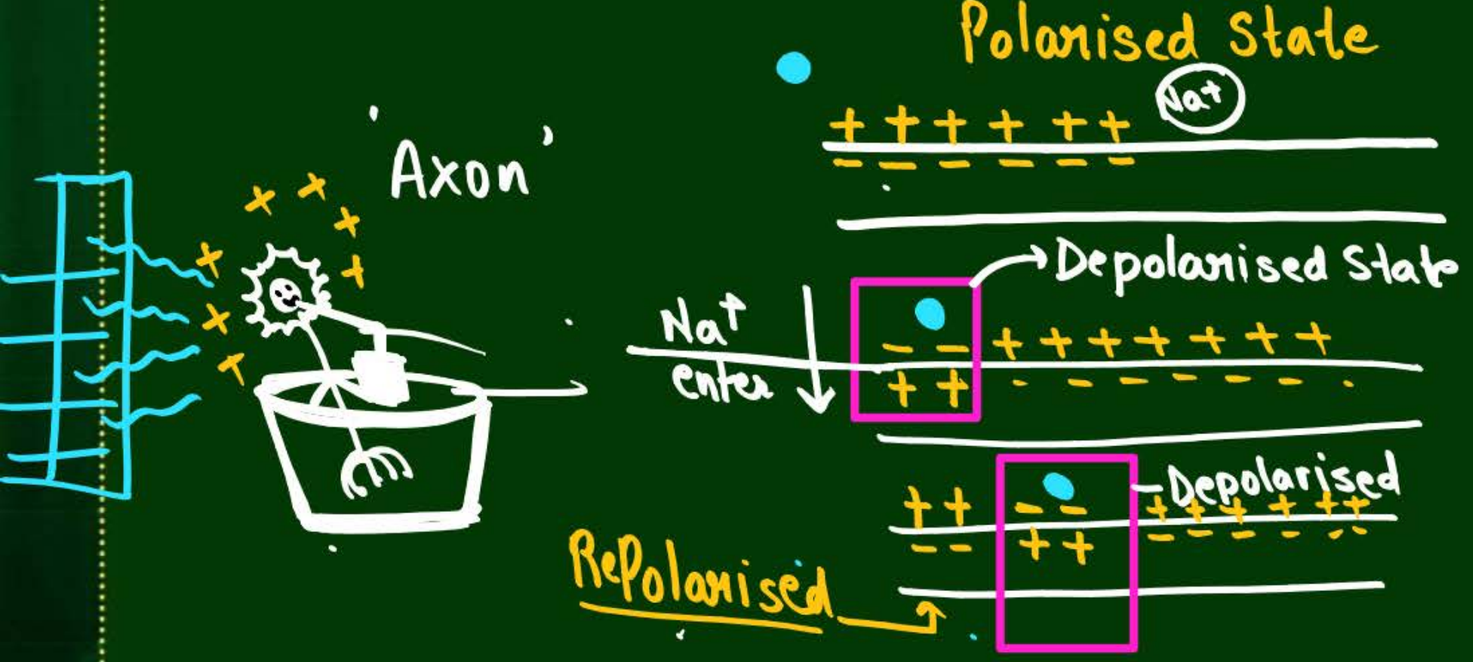


IMPORTANT TERMS IN NERVOUS ACTIVITY

Impulse :

- A wave of irritability (electrical disturbance that sweeps over the nerve cell).

① More Na^+ ions are present outside the cell. (Resting State)



ACTION POTENTIAL

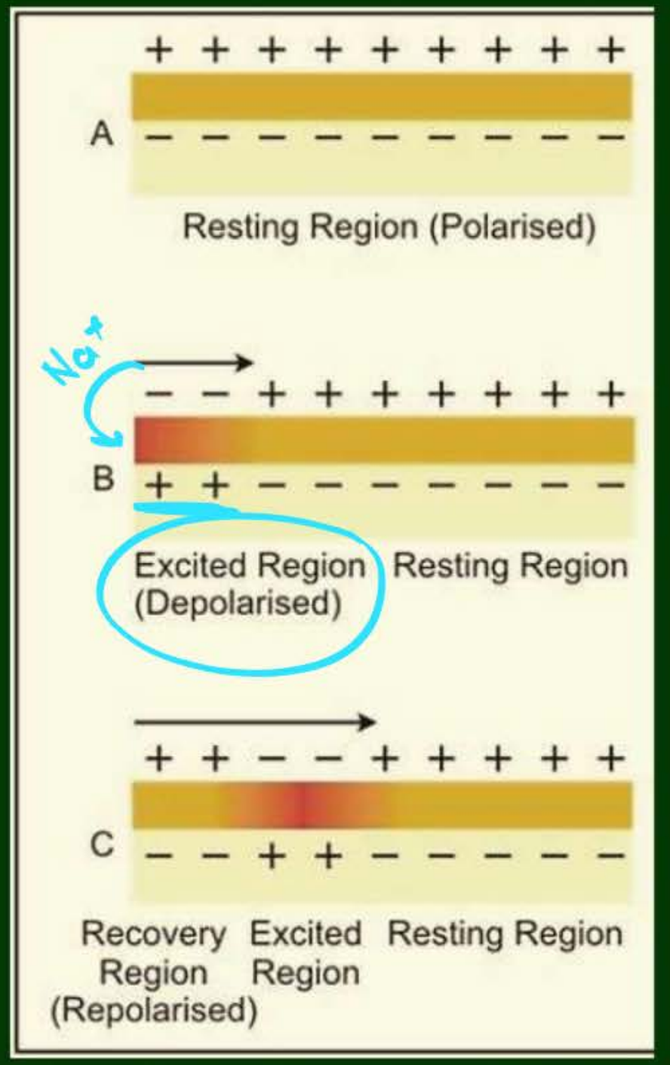
Polarised state

- In the normal (resting) condition, the outer side of the nerve fibre carries positive (+) charge.
- This polarisation is due to more Na^+ ions outside the axon membrane.

Depolarised state

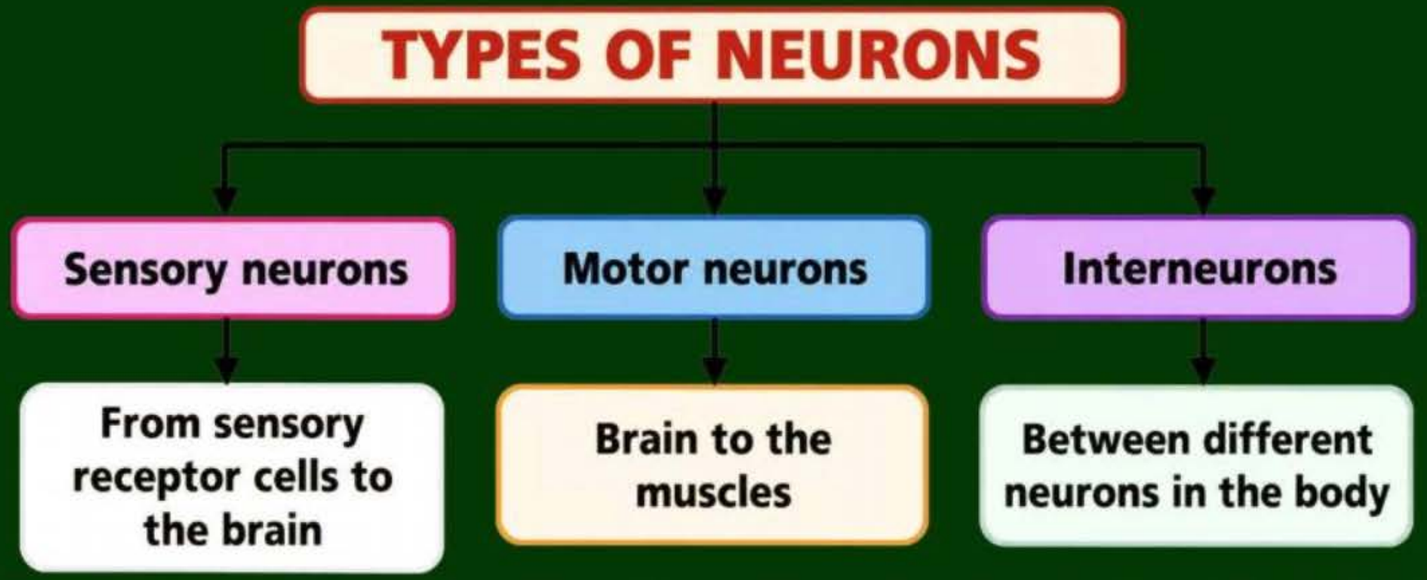
touch (movement of ions)

- On stimulation (mechanical, electrical, chemical or heat, etc.), the axon membrane at that spot becomes more permeable to Na^+ ions which move inwards and cause loss of polarisation (depolarisation). This is known as the excited region. (Depolarisation)
- The point of depolarisation becomes a stimulus for the next neighbouring area of the membrane which in turn becomes depolarised.



TYPES OF NEURONS

- **Sensory neurons** convey the impulse from the receptor (sense organ) to the main nervous system (the brain or spinal cord).
- **Motor neurons** carry the impulse from the main nervous system to an effector (muscle or gland).
- **Association (connecting) neurons** are located in the brain and spinal cord, which interconnect the sensory and motor neurons.



TYPES OF NEURONS

SIGNALS PASS THROUGH AND BETWEEN NEURONS

↓

Information is collected by the dendritic tips

↓

The chemical reaction sets off electric impulse

↓

Electric impulses pass from dendrite to cell body

↓

Then passes from cell body through the axon

↓

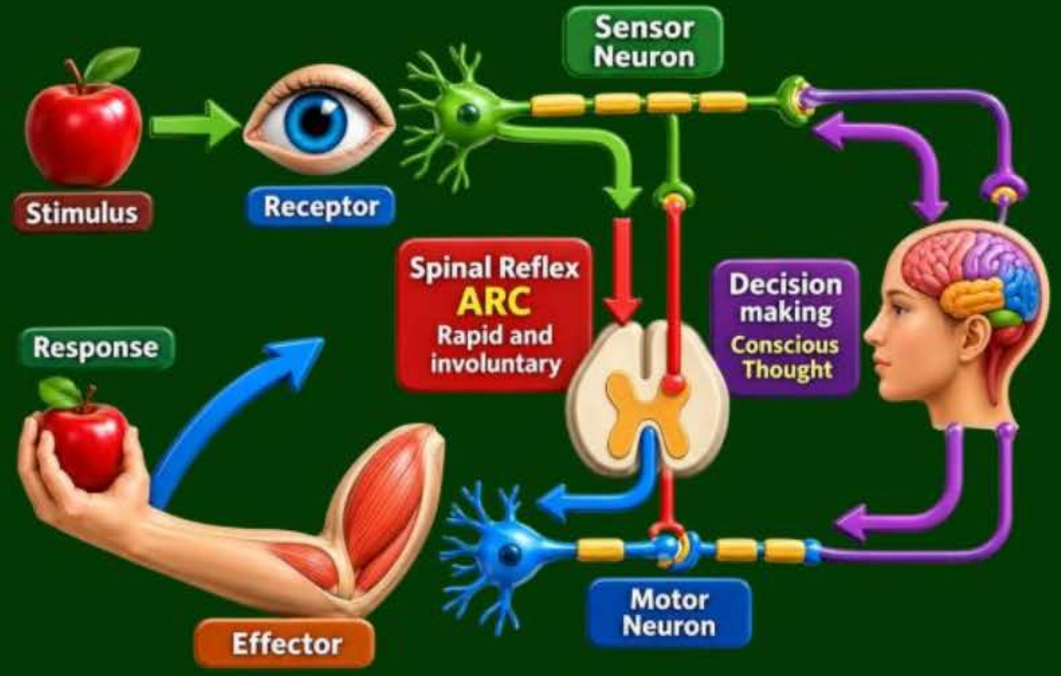
Reaches nerve endings at the end of axon

↓

Electric impulse sets off signals called neurotransmitters across the synapse

↓

This again passes as electric signals in the next dendrite and so on



QUESTIONS

The basic structural and functional unit of brain is

A Cyton

B Nephron

C Axon

D Neuron ✓

QUESTIONS

Which of the following structures is absent in nerve cells?

A. Nucleus

B. Cytoplasm

C. Centrosome

D. Neurilemma

QUESTIONS

Nissl's granules are found in the cytoplasm of:

A Nerve cell

B Dendrons

C Axon

QUESTIONS

The Myelin sheath is derived from the

A Microglia

B Neuroglial cells

C Schwann cells

D Nerve cells

Thank You